Hamilton-Jacobi Reachability

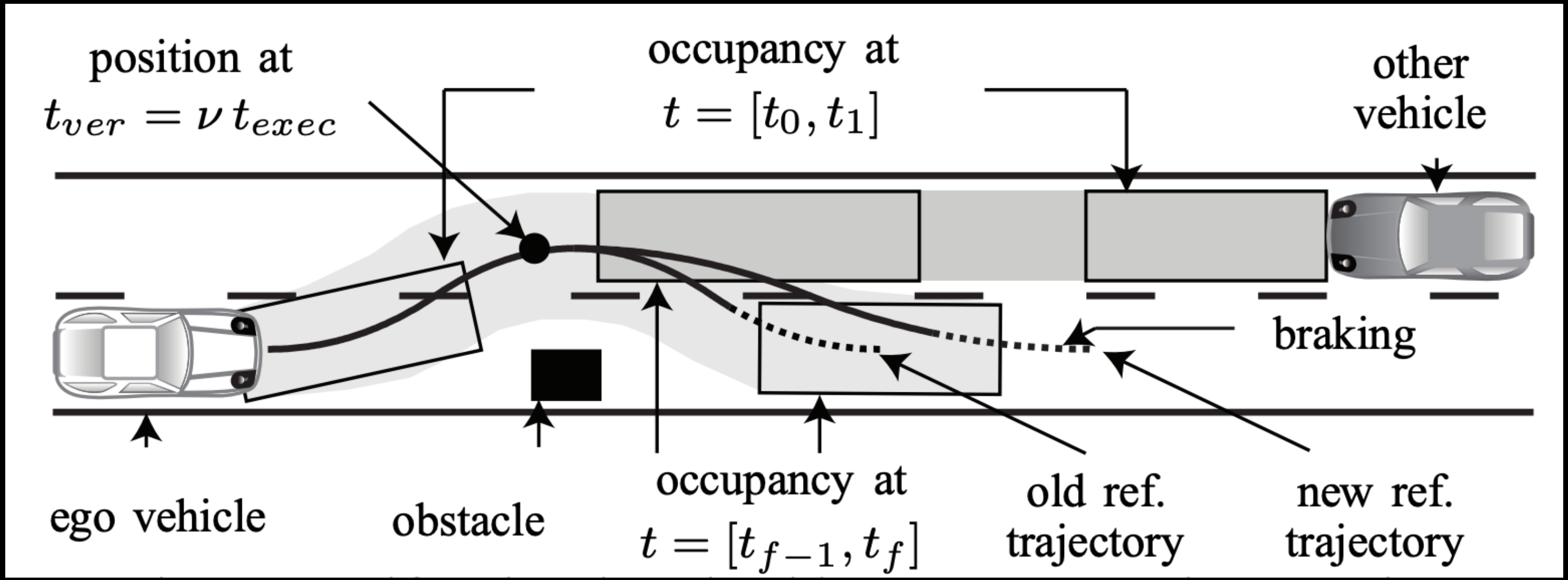
Reachability analysis

- The study of the set of states a dynamical system can (and can't) reach over time given:
  - dynamics
  - initial conditions,
  - possible controls
  - disturbances

Reachability analysis: Applications

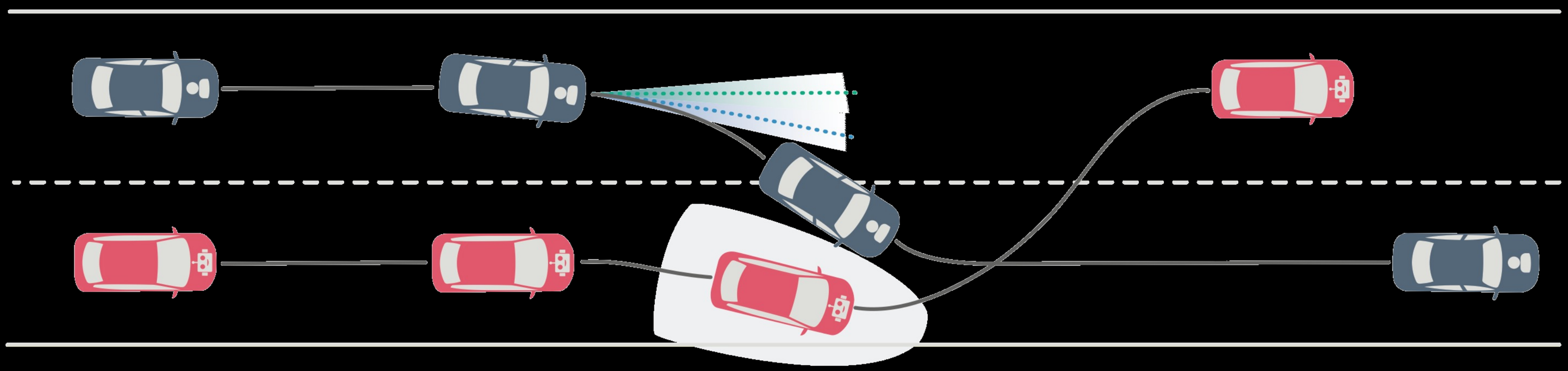
- Safety-critical control: Avoid inevitable collision states
- Robust control: Plan a "tube path" to avoid obstacles
- Provide guarantees: Certify safe/unsafe states before deployment

Reachability analysis: Applications



M. Althoff and J. M. Dolan, "Online Verification of Automated Road Vehicles Using Reachability Analysis," in *IEEE Transactions on Robotics*, vol. 30, no. 4, pp. 903-918, Aug. 2014,

Reachability analysis: Applications



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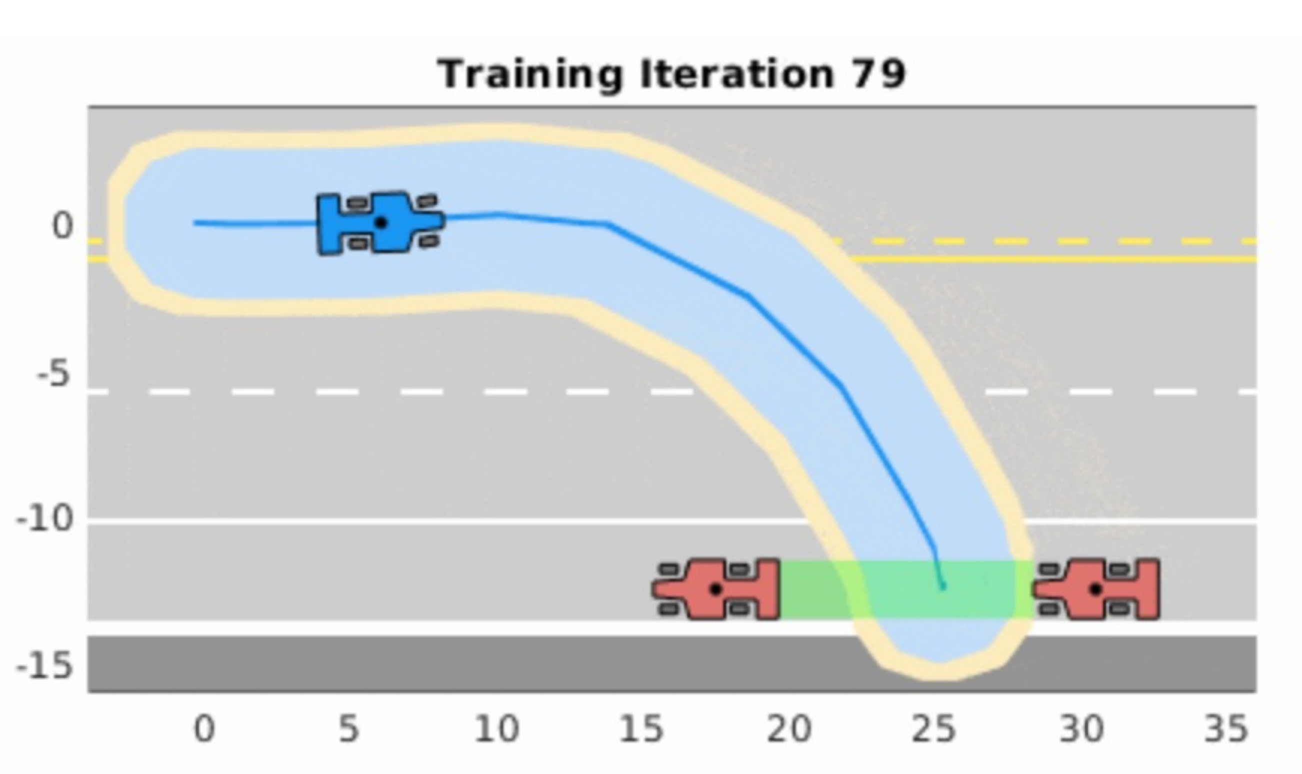
Reachability analysis: Applications



Forward reachability (open-loop)

Forward reachability: The set of states a system can reach within some time using any control input

"Starting from here, under any allowed input/disturbance, where could we end up?"

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"Starting from here, under any allowed input/disturbance, where could we end up?"

Open loop because:

- Assume all possible control inputs are applied without correction along the way. I.e., no feedback.

Backward reachability (closed-loop)

Backward reachability: Given a *target* set, what are all the states that could reach (or avoid) that set in the future?

"Where must I be now to reach the target set later?"

"Where must I avoid now to avoid the target set later?"

Backward reachability (closed-loop)

Backward reachability: Given a *target* set, what are all the states that could reach (or avoid) that set in the future?

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"Where must I avoid now to avoid the target set later?"

Closed-loop because

- At each state and time, we consider the best control to take to reach/avoid the target set

Sounds like dynamic programming…

Backward reachable sets

Reach case

$$\mathcal{R}(\mathcal{T}, -t) = \{x_0 \in \mathcal{X} \mid \exists u(\cdot) \in \mathbb{U}[-t, 0]\,\text{s.t.}\ \xi_{x_0,-t}^{u(\cdot)}(0) \in \mathcal{T}\}$$

Avoid case (Inevitable collision set)

$$\mathcal{A}(\mathcal{T}, -t) = \{x_0 \in \mathcal{X} \mid \forall u(\cdot) \in \mathbb{U}[-t, 0]\,\text{s.t.}\ \xi_{x_0,-t}^{u(\cdot)}(0) \in \mathcal{T}\}$$

How to compute BRS?

- This is an optimal control problem! But we only care about whether we reach/avoid target set at the **end** of the horizon

$$
\begin{aligned}
\min_{u(\cdot)\in\mathbb{U}[0,T]} \quad & \int_0^T g(x,u,t)dt + g_T(x(T)) \\
\text{subject to} \quad & \dot{x} = f(x,u,t) \\
& x_0 = x(0) \\
& u \in \mathcal{U}(x), x \in \mathcal{X}
\end{aligned}
$$

Define target set & value function interpretation

Consider reach case. Being inside target set is desirable

$$\mathcal{T} = \{x \in \mathcal{X} \mid g_T(x) \leq 0\}$$

Cost-to-go

$$V(x,t) = \min_{u(\cdot)\in\mathbb{U}[t,T]} g_T(x(T))$$

$$V(x,-t) = \min_{u(\cdot)\in\mathbb{U}[-t,0]} g_T(x(0))$$

Do the optimal thing to get inside target set as far as possible

What about the **avoid** case?

Going back to HJB equation (BRS)

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \{g(\boldsymbol{x}(t), \boldsymbol{u}(t), t) + \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t)\} = 0$$

Terminal condition: $V(x, T) = g_N(x)$

Zero running cost!

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t) = 0$$

Solve this
PDE…

Wait a minute…what if trajectories *enter* the target set, but then escape it afterwards?

$$V(x,-t) = \min_{u(\cdot)\in\mathbb{U}[-t,0]} \left(\min_{s\in[-t,0]} g_T(x(s)) \right)$$

$$\mathcal{T} = \{x \in \mathcal{X} \mid g_T(x) \le 0\}$$

$$V(x,t) = \min_{u(\cdot)\in\mathbb{U}[0,t]} \left(\min_{s\in[0,t]} g_T(x(s)) \right)$$

Lowest cost over horizon

Backward reachable tube

Reach case

$$\widetilde{\mathcal{R}}(\mathcal{T}, -t) = \{x_0 \in \mathcal{X} \mid \exists u(\cdot) \in \mathbb{U}[-t, 0],\ \exists s \in [-t, 0]\,\text{s.t.}\ \xi_{x_0,-t}^{u(\cdot)}(s) \in \mathcal{T}\}$$

Avoid case (Inevitable collision set)

$$\widetilde{\mathcal{A}}(\mathcal{T}, -t) = \{x_0 \in \mathcal{X} \mid \forall u(\cdot) \in \mathbb{U}[-t, 0],\ \exists s \in [-t, 0],\ \text{s.t.}\ \xi_{x_0,-t}^{u(\cdot)}(s) \in \mathcal{T}\}$$

Going back to HJB equation (BRT)

BRS:

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t) = 0$$

BRT:

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min(0, \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t)) = 0$$

Idea: Find lowest value. So we can “freeze” value ($\frac{\partial V}{\partial t} = 0$) if system is moving *away* from target set ($\min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t) \geq 0$)

Summary so far

- Backward reachability useful for *safety analysis*
  - Is it possible to reach/avoid a set within some time?
- We can use the HJB equation to help us solve some backward reachability problems
  - Zero running cost. Value function has a clear interpretation.
- Reach case:

BRS:

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t) = 0$$

BRT:

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min(0, \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t)) = 0$$

- Avoid case: $\max_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))}$

What about disturbances?

- Consider disturbances in the dynamics $\dot{x} = f(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{d}, t)$
- We would like to be ***robust*** against any and all disturbances
  - Can we still **reach** the target set despite worst-case disturbances?

What about disturbances?

- Consider disturbances in the dynamics $\dot{x} = f(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{d}, t)$
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    - Can we still **avoid** target set despite worst-case disturbances?

What about disturbances?

Perform reachability analysis assuming ***adversarial*** disturbances

- If control wants to *reach* target, then disturbance aims to *avoid* target
- If control wants to *avoid* target, then disturbance aims to *reach* target

With disturbances

Reach case

$$\mathcal{R}(\mathcal{T}, -t) = \{x_0 \in \mathcal{X} \mid \forall d(\cdot) \in \mathbb{D}[-t, 0], \exists u(\cdot) \in \mathbb{U}[-t, 0] \text{ s.t. } \xi_{x_0,-t}^{u(\cdot)}(0) \in \mathcal{T}\}$$

$$\widetilde{\mathcal{R}}(\mathcal{T}, -t) = \{x_0 \in \mathcal{X} \mid \forall d(\cdot) \in \mathbb{D}[-t, 0], \exists u(\cdot) \in \mathbb{U}[-t, 0], \exists s \in [-t, 0], \text{s.t.} \xi_{x_0,-t}^{u(\cdot)}(s) \in \mathcal{T}\}$$

Avoid case

$$\mathcal{A}(\mathcal{T}, -t) = \{x_0 \in \mathcal{X} \mid \exists d(\cdot) \in \mathbb{D}[-t, 0], \forall u(\cdot) \in \mathbb{U}[-t, 0] \text{ s.t. } \xi_{x_0,-t}^{u(\cdot)}(0) \in \mathcal{T}\}$$

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Ordering of control and disturbances

- Consider BRS-reach

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t) = 0$$

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \max_{\boldsymbol{d}(t) \in D(\boldsymbol{x}_t)} \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{d}(t), t) = 0$$

or

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \max_{\boldsymbol{d}(t) \in D(\boldsymbol{x}_t)} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{d}(t), t) = 0$$

- Which one should we pick?

Max-min inequality

For any function $f: Z \times W \to R$,

$$\sup_{z \in Z} \inf_{w \in W} f(z, w) \leq \inf_{w \in W} \sup_{z \in Z} f(z, w)$$

- **Interpretation**

$$\sup_{z \in Z} \inf_{w \in W} f(z, w)$$

Player 1 **Player 2**

- **Takeaway**
  - Acting second has the advantage!

HJ Reachability with disturbances

- **BRS, reach, with disturbances**

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \max_{\boldsymbol{d}(t) \in D(\boldsymbol{x}_t)} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{d}(t), t) = 0$$

- **BRT, reach, with disturbances**

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min\left(0, \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \max_{\boldsymbol{d}(t) \in D(\boldsymbol{x}_t)} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{d}(t), t)\right) = 0$$

- **BRS, avoid, with disturbances**

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \max_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \min_{\boldsymbol{d}(t) \in D(\boldsymbol{x}_t)} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{d}(t), t) = 0$$

- **BRT, avoid, with disturbances**

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min\left(0, \max_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \min_{\boldsymbol{d}(t) \in D(\boldsymbol{x}_t)} \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{d}(t), t)\right) = 0$$

What if we have control-disturbance-affine dynamics?

- Suppose we have dynamics $\dot{\boldsymbol{x}} = f_0(\boldsymbol{x}) + B_u(\boldsymbol{x})\boldsymbol{u} + B_d(\boldsymbol{x})\boldsymbol{d}$
- Assume control and disturbances are *bounded*
    - $\boldsymbol{u} \in [\boldsymbol{u}_{\min}, \boldsymbol{u}_{\max}]$, $\boldsymbol{d} \in [\boldsymbol{d}_{\min}, \boldsymbol{d}_{\max}]$
- Consider the term $\nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{d}(t), t))$
- Is solving for the optimal control/disturbance tractable?

Computing HJ reachable sets

- https://github.com/StanfordASL/hj_reachability
- https://github.com/StanfordASL/hj_reachability/blob/main/examples/quickstart.ipynb

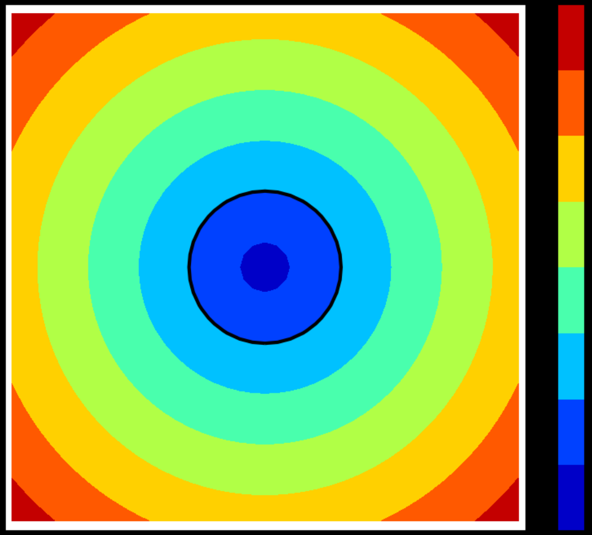
```
dynamics = hj.systems.Air3d()

grid = hj.Grid.from_lattice_parameters_and_boundary_conditions(hj.sets.Box(np.array([-6., -10.,
0.]), np.array([20., 10., 2 * np.pi])), (51, 40, 50), periodic_dims=2)

values = jnp.linalg.norm(grid.states[..., :2], axis=-1) - 5

solver_settings = hj.SolverSettings.with_accuracy("very_high",
hamiltonian_postprocessor=hj.solver.backwards_reachable_tube)
```

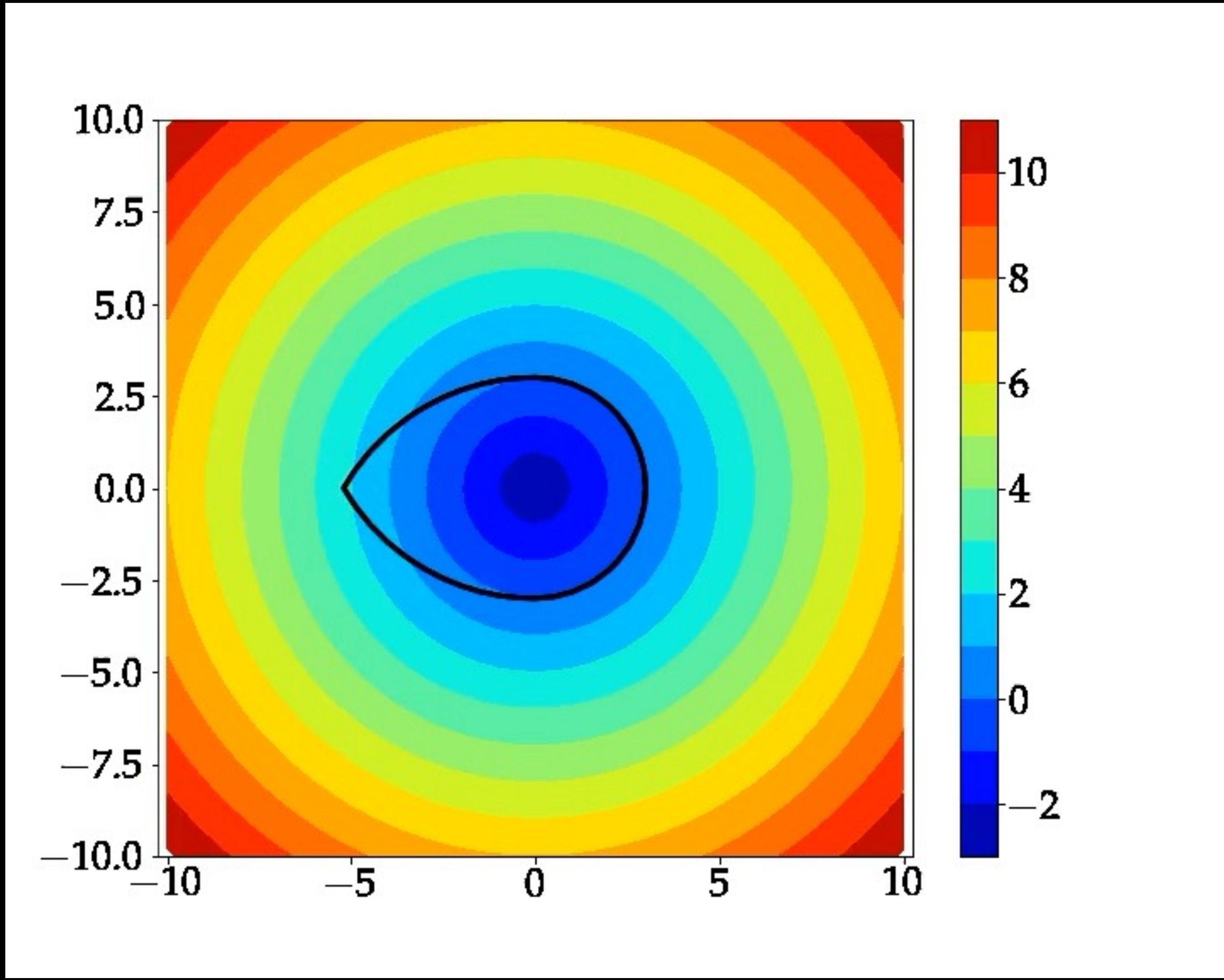
Dubins car example (BRT, avoid)

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \\ u + d \end{bmatrix}$$

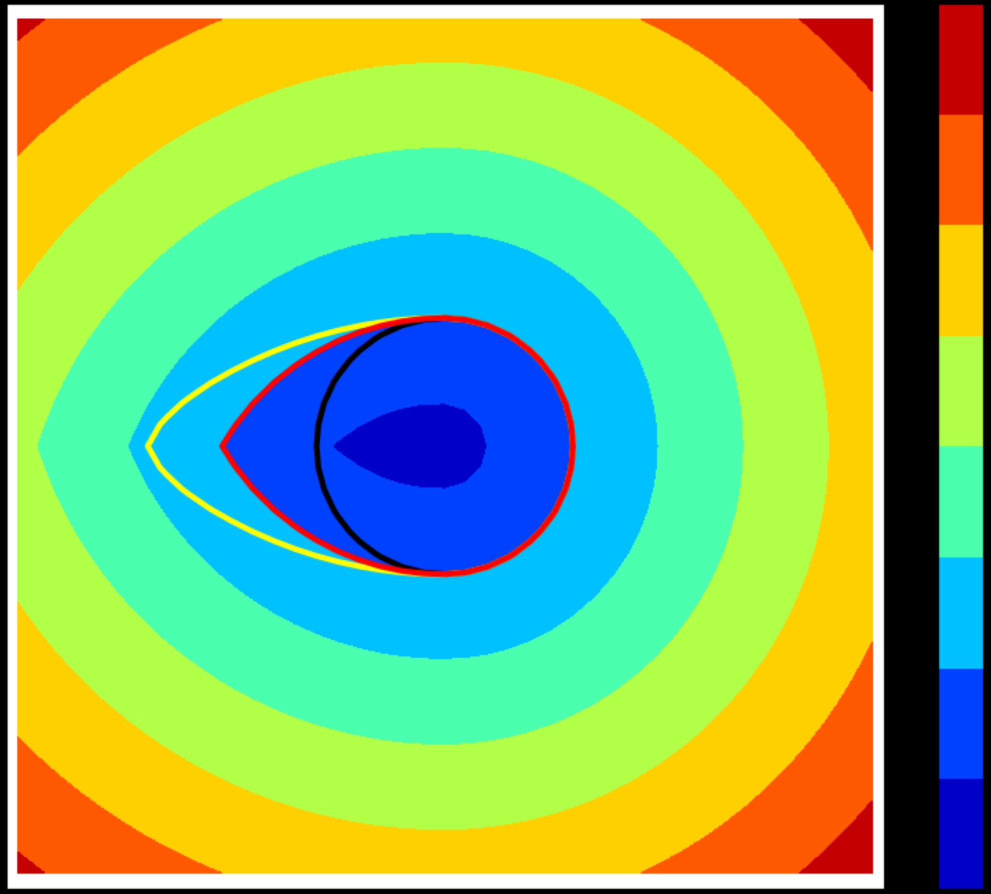
Constant velocity

Can only control yaw rate





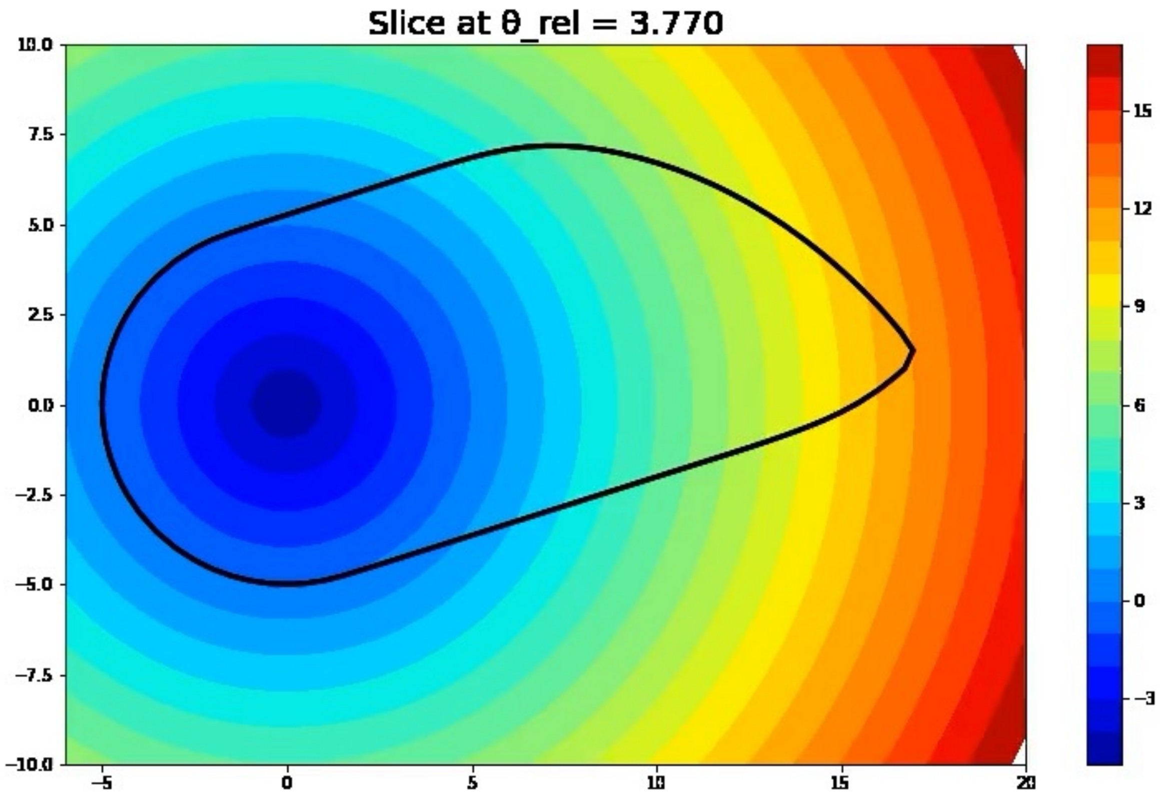
Dubins car example with disturbances

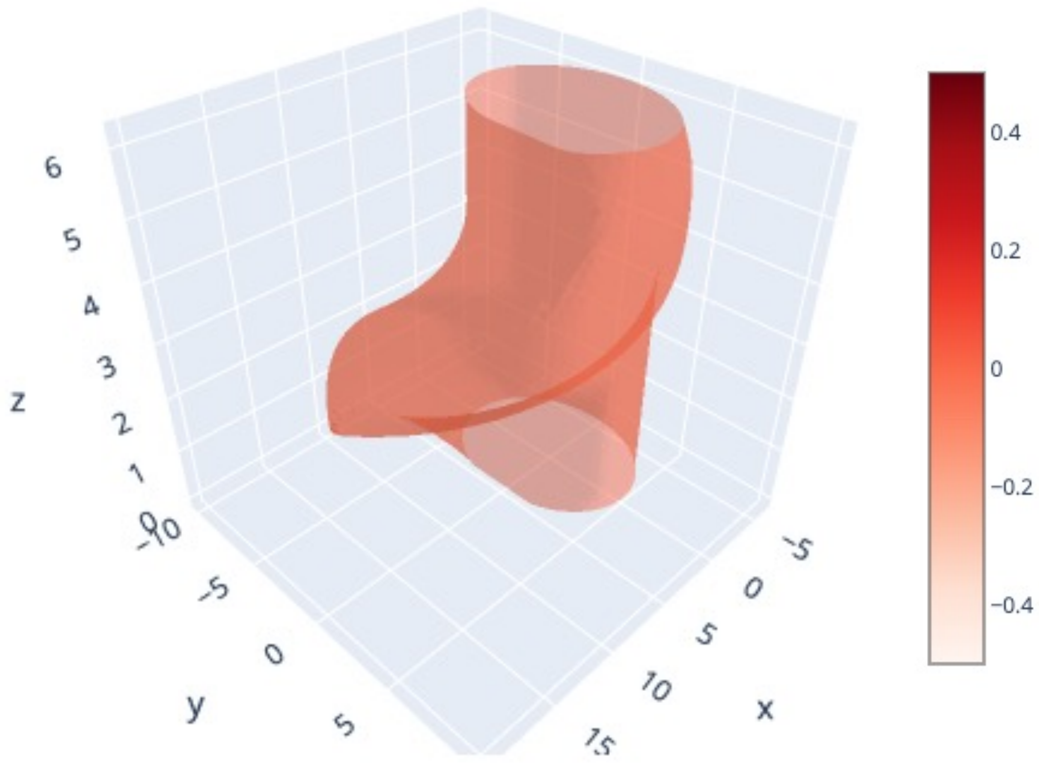


Pursuit-evasion example

- Can consider a *relative* system between two agents & treat other agent as a disturbance
- Consider two Dubins' car agents, one is pursuing the other (i.e., cat and mouse)

Pursuit-evasion example

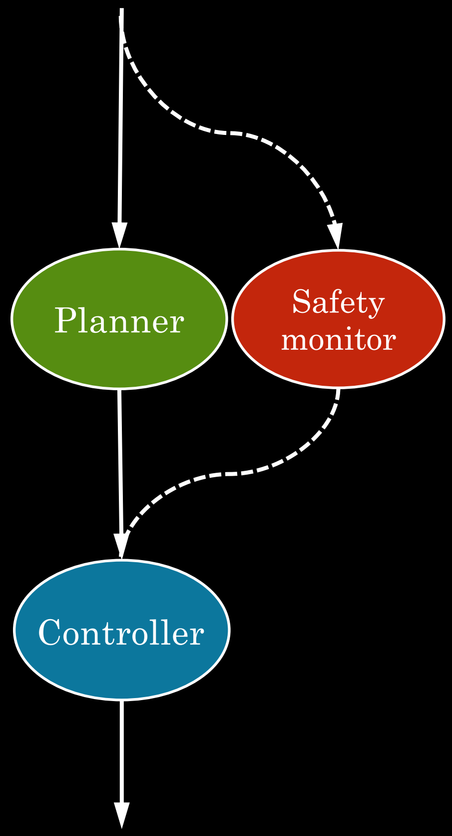




What can I do with the HJ value function?

- HJ value functions are CBFs!
  - https://arxiv.org/abs/2104.02808
  - https://youtu.be/9SIO-oNT7Gs
- Least-restrictive controller
  - When $V(x,t) \leq \epsilon$, switch to optimal HJ policy
- Minimally-interventional safe controller
  - When $V(x,t) \leq \epsilon$, add HJ control constraint in tracking controller
  - https://arxiv.org/abs/2012.03390

Minimally-interventional control



Minimize

Tracking error

Subject to

Control limits

Dynamics

State constraints

+

If $\boldsymbol{V(x,t) < \epsilon}$
then activate HJ safety-preserving control set constraint

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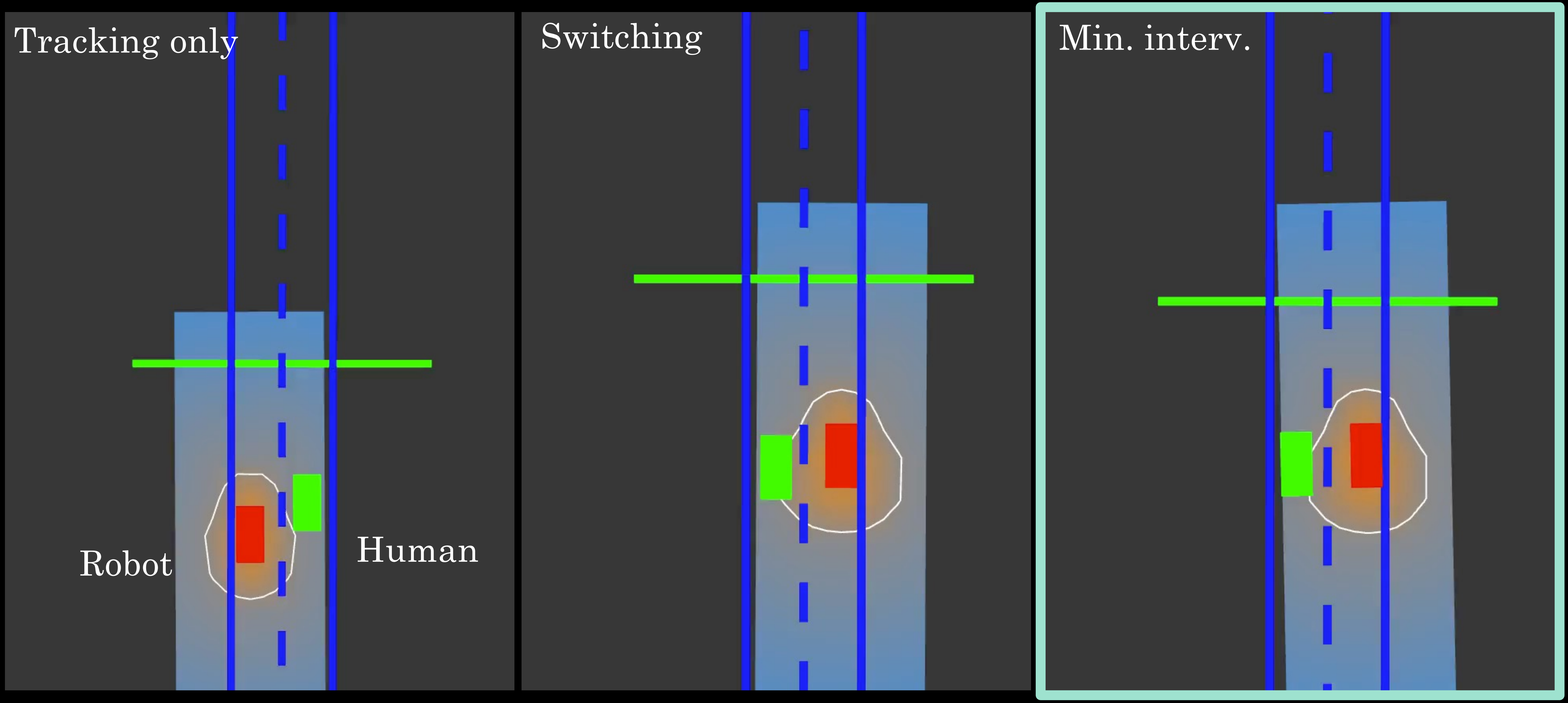
Tracking only
Robot
Human
Switching
Min. interv.